

Mutation-Aware Docking Platform — Concept Document

The Idea in One Sentence

A web platform where users pick a protein mutation and a set of compounds, and the system automatically generates the mutant structure, docks the compounds against both wild-type (WT) and mutant, and returns a clear visual comparison showing whether each compound prefers the mutant — all without the user touching PyMOL, FoldX, or AutoDock manually.

Why This Is Different From What Exists

Free docking servers like SwissDock, CB-Dock2, HDock, and MTiOpenScreen are widely used (SwissDock alone has ~530,000 users and ~710,000 dockings since 2011). But none of them solve the workflow that mutation-focused researchers actually want:

- **No mutation handling.** Users must apply mutations manually in PyMOL, FoldX, or Rosetta before uploading.
- **No WT-vs-mutant comparison.** Users run two separate jobs and compare scores by hand in Excel.
- **No selectivity matrix.** No tool lets you dock N compounds against M mutations and see a sortable matrix.
- **Weak visualization.** Basic 3D viewers, no interaction fingerprints, no clash analysis, no plain-English interpretation.
- **Bare-bones batch UX.** Where bulk is supported, results come back as ZIP files, not a usable interface.

The opportunity is to reframe the product from *“a docking server”* to *“a mutation-selectivity analysis tool.”*

Target Users

Most likely to pay, ranked:

1. **Small biotechs** without Schrödinger licenses, doing early-stage discovery.
2. **Educators** teaching structure-based drug design — institutional licenses for teaching labs.
3. **Consultants and freelance med-chemists** doing one-off projects.
4. **Power-user academics** who hit free server quotas or need API access.

Won't pay: big pharma (have internal pipelines, won't trust external servers with proprietary compounds), serious comp-chem groups (run their own Vina/Glide), most individual academics (will use SwissDock for free).

Architecture & Tech Stack

Frontend

- React or Vue
- Mol* or NGL Viewer for 3D
- Ketcher or JSME for ligand drawing
- ProtVista or custom D3 component for clickable mutation selection on protein sequence

Backend

- FastAPI (Python, async)
- Celery + Redis or RQ for job queueing
- PostgreSQL for jobs/results
- S3-compatible object storage for structures and ligand files

Compute

- RunPod for CPU spot instances (Vina is CPU-bound, very cheap)
- Dockerized environments
- Cap users at small jobs initially; scale based on demand

Docking Engine

- AutoDock Vina (default) — free, fast, scriptable

- smina as alternative (better scoring options)
- DiffDock or Uni-Mol for ML-based blind docking (optional, GPU-hungry)

Supporting Tools

- RDKit — SMILES → 3D conformer
 - Open Babel — format conversion
 - Meeko — Vina-compatible PDBQT prep
 - PDBFixer — clean structures
 - ProLIF or PLIP — interaction fingerprints
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Mutation Modeling Pipeline

Tier 1 — Side-chain swap (fast, surface mutations)

- PyMOL mutagenesis wizard (scriptable)
- SCWRL4 for fast side-chain placement

Tier 2 — Local refinement (pocket-affecting mutations)

- FoldX `BuildModel` (sweet spot: seconds, gives $\Delta\Delta G$)
- Rosetta `cartesian_ddg` or `relax` (slower, more accurate)
- Short MD relaxation in OpenMM (most realistic)

Tier 3 — No structure available

- ESMFold (fast) or AlphaFold2/3 (more accurate) on mutant sequence
- Note: AlphaFold is often insensitive to point mutations — use it for the fold, then FoldX for the local effect

Tier 4 — Backbone-disrupting mutations

- Rosetta remodel, loop modeling, or full MD
- Out of scope for v1 — flag as “advanced” or unsupported

Default pipeline: Fetch WT structure → ESMFold if missing → FoldX mutation → optional OpenMM minimization → dock

End-to-End Data Flow

1. **Target selection** — UniProt ID + mutation (e.g., EGFR T790M)
2. **WT fetch & prep** — PDB download → PDBFixer cleanup
3. **Mutation application** — FoldX BuildModel → mutant.pdb + $\Delta\Delta G$
4. **Pocket definition** — fpocket / P2Rank or co-crystal ligand centroid
5. **Ligand prep** — SMILES → RDKit 3D conformer → Meeko → PDBQT
6. **Receptor prep** — Meeko → receptor.pdbqt
7. **Docking** — Vina with pocket box (30s–3min per ligand)
8. **Post-processing** — pose conversion, ProLIF interaction analysis, optional MM-GBSA rescoring
9. **Results to frontend** — JSON + presigned URLs to pose files

Always run WT alongside mutant by default. The Δ -score is what users actually want.

Caching wins (in priority order):

- Prepared mutant structures
- Ligand PDBQTs (canonical SMILES key)
- WT prepped structures
- Full docking results keyed on (mutant_hash, ligand_hash, pocket_hash)

Differentiating Features (The Real Wedge)

Selectivity matrix

A 2D table users actually want:

Compound	WT	T790M	L858R	C797S
Gefitinib	-7.2	-5.1	-8.4	-4.9
Osimertinib	-8.8	-9.3	-8.9	-5.2
Compound X	-6.1	-7.8	-6.5	-7.1

Color-coded, sortable, clickable to drill into 3D pose. Filter by "binds mutant better than WT." Export to CSV.

Plain-English interpretation

Instead of "Vina score: -8.2," surface: *"Compound X is predicted to bind 1.2 kcal/mol better to the mutant than WT, primarily through new contact with M790."*

Synced 3D visualization

- Mutant pocket and WT pocket side by side, synced rotation
- Mutation residue highlighted, contacts colored by interaction type (H-bond green, hydrophobic yellow, clash red)
- Slider to morph between WT and mutant
- 2D interaction diagrams via ProLIF
- Clash report: *"M790 side chain comes within 2.1 Å of ligand piperazine — predicted clash"*

One-click clinical mutation library

Pre-curated entries for clinically important mutations: EGFR T790M / C797S / L858R, KRAS G12C / G12D, BRAF V600E, IDH1 R132H, ABL T315I, etc. User picks from a list, no setup needed.

Business Viability

Market size signals

- SwissDock: ~150 dockings/day, ~40,000 new users/year
- CB-Dock2: 200+ submissions/day
- HDock: 30,000+ jobs since 2017
- HADDOCK: 17,000+ citations

Demand is real. The challenge: free services set the price ceiling at zero for basic docking.

Pricing models that could work

- Freemium: 5 free dockings/month, then pay
- Per-project: \$20–100 for a defined screening campaign

- API access for developers: \$/call
- Institutional licenses: \$500–5,000/year per lab

Cost reality

- Vina is CPU-bound; RunPod CPU spot ~\$0.10–0.20/hr
- 200 docking runs (50 compounds × 4 structures) ≈ pennies of compute
- Customer acquisition is the harder cost, not compute

Conservative math

- 1% conversion of SwissDock-scale users at \$50/year = \$20,000/year
 - 5% conversion at \$200/year = \$400,000/year
 - Both plausible if differentiation is genuine
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Realistic Risks

- **Mutation modeling quality** — naive side-chain swaps mislead for buried/backbone-affecting mutations. Need to be transparent about confidence.
 - **Docking score interpretation** — absolute scores are unreliable; users may misinterpret “this drug works.” Δ-scores and clear UI framing help.
 - **IP concerns** — biotechs won’t upload proprietary compounds without strong privacy guarantees. Consider in-browser docking option (like MolModa).
 - **Free tool competition** — must be meaningfully better, not marginally. Mutation-aware workflow is the wedge.
 - **Customer acquisition** — scientific software has brutal CAC; community-first growth (free useful tool → paid features) usually beats day-one paywall.
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MVP to Test Demand Cheaply

Before building full RunPod infrastructure:

1. Mock up the selectivity matrix UI as an interactive HTML demo
2. Pre-compute results for 5–10 clinically important mutations × 20 known compounds

3. Put it behind a waitlist or email capture
4. Share in relevant communities (biotech Twitter/X, LinkedIn, Reddit r/bioinformatics, lab Slacks)
5. See who signs up and what they ask for

If 100+ relevant people sign up in a few weeks, build the real backend. If 10, rethink the wedge.

Domain Name Candidates

Mutation-selectivity focused (top picks)

- MutaDock
- DeltaDock
- SelectiBind / Selectiv
- MutantFit
- MutaScore
- VariantBind

Speed/ease angle

- QuickDock / SnapDock / FlashDock
- DockNow
- OneDock

Clinical / precision medicine

- PrecisionDock
- TargetShift
- ResistDock
- ClinDock

Brandable / abstract

- Bindly

- Pocketly
- Dockwise
- Affyn / Affynix
- Molyx
- Kindock
- Aposite

Top 3 picks

1. **MutaDock** — clearest positioning, speaks directly to differentiator
2. **DeltaDock** — captures Δ -score concept, technical-cool, niche
3. **Bindly** — brandable SaaS-style, room to grow beyond docking

Domain extensions to consider

- `.bio` — signals biotech credibility, great fit
- `.io` — startup default
- `.science` — cheap and descriptive
- `.com` — only if available at normal registration; avoid premium parking fees

Verify availability on Namecheap, Porkbun, or Cloudflare Registrar before committing.

Summary

The idea is technically straightforward and the demand is unambiguously real. The hard parts are differentiation (mutation-aware workflow + WT-vs-mutant matrix + visual interpretation are the wedge) and customer acquisition (free competitors set the ceiling — must be meaningfully better, not marginally). Compute economics are favorable on RunPod. Pricing should target small biotechs, educators, and consultants — not individual academics. Best path forward: build a cheap MVP demo to validate demand before committing to full infrastructure.